

Box Models

From Ch16-18, we are going to show to use a unified way — through a **box model**, to analyze a huge class of **chance processes**, like

- ▶ The number of squares moved in a play at monopoly.
- ▶ The number of heads in 10,000 tosses of a coin.
- ▶ The amount of money won or lost gambling at roulette.
- ▶ The number of Democrats in a random sample of voters.

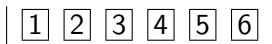
The Sum of The Draws

Many chance processes are like drawing from a box, that the chance quantity of interest is like the sum of numbers drawn at random from a box.

- ▶ There is a box of tickets.
- ▶ Each ticket has a number written on it.
- ▶ Some tickets are drawn at random from the box.
 - ▶ Typically with replacement, and at each draw, all the tickets in the box are equally likely to be drawn
- ▶ The numbers on the tickets are added up.
 - ▶ The total is called the **sum of the draws**.

Example 1: Rolling a pair of dice

The total number spots resulting from rolling a pair of dice is like the sum of _____ draws from the box

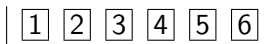


- ▶ On a single roll of the die, any one of the numbers 1, 2, ..., 6 could come up. So the tickets in the box should be labeled 1, 2, 3, 4, 5, and 6.
- ▶ Each number is equally likely, so there should be equal number of tickets of each kind.
- ▶ Rolling a pair of dice is like making _____ draws from the box.
- ▶ To illustrate the analogy:

Rolls of the dice:	 	Total number of spots: 8
Draws from the box:	3 5	Sum of the draws: 8

Example 1: Rolling a pair of dice

The total number spots resulting from rolling a pair of dice is like the sum of 2 draws from the box

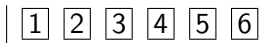


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Rolls of the dice:			Total number of spots: 8
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Example 2: Tossing a coin

The number of heads in 5 tosses of a fair coin is like the sum of _____ draws from the box

$\boxed{0} \boxed{1}$ where $\boxed{1} \rightarrow \text{head}$, and $\boxed{0} \rightarrow \text{tail}$.

- ▶ Think about counting the number of heads in the 5 tosses.
On each toss:
 - ▶ you add 1 to the count, if you get a head;
 - ▶ you add 0 to the count, if you get a tail.

So the tickets in the box should be labeled $\boxed{1}$ and $\boxed{0}$.

- ▶ Heads and tails are equally likely, so there should be equal number of ticket of each kind.
- ▶ There should be _____ draws.
- ▶ To illustrate the analogy:

Tosses: H T H H T

Number of heads: 3

Draws: $\boxed{1} \boxed{0} \boxed{1} \boxed{1} \boxed{0}$

Sum of the draws: 3

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Number of heads: 3

Draws: $\boxed{1} \boxed{0} \boxed{1} \boxed{1} \boxed{0}$

Sum of the draws: 3

Suppose the coin in the previous example is **not fair**, but slightly biased in favor of heads — it lands heads 51% of the time, and tails the other 49%. How would the box model have to be changed?



In making a box model, what three questions need to be answered?

- ▶ What numbers go into the box?
- ▶ How many of each kind?
- ▶ How many draws?

How do you answer the first two questions?

- ▶ What numbers the chance process can possibly generate,
- ▶ What are the chances for each number. You can find this by thinking about how many ways there are to generate each number.

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51 tickets	1	49 tickets	0
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Example 3: Roulette (FPP, page 282)

A gambler is going to play roulette 5 times, putting a dollar on 11 and 12 each time. If either 11 or 12 comes up, the gambler gets the dollar back, together with winnings of \$17. If neither number comes up, he loses his dollar.

0	00	1	2	...	10	11	12	13	...	36
↓	↓	↓	↓	...	↓	↓	↓	↓	...	↓
-\$1	-\$1	-\$1	-\$1	...	-\$1	\$17	\$17	-\$1	...	-\$1

The gambler's net gain in the 5 plays is like the sum of _____ draws from the box

\$17	\$17	36 tickets	-\$1
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To illustrate the analogy:

Spin:	5	11	00	36	2	
Win/Loss:	-\$1	\$17	-\$1	-\$1	-\$1	Net gain: \$13
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Chuck-a-luck



In one version of chuck-a-luck, 3 dice are rolled out of a cage. You can bet that all 3 show . The house pays 180 to 1, and the bettor has ____ chance in ____ to win.

Suppose you make this bet 10 times, staking \$1 each time.

Your net gain is like the sum of ____ draws made at random with replacement from the box

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In one version of chuck-a-luck, 3 dice are rolled out of a cage. You can bet that all 3 show . The house pays 180 to 1, and the bettor has 1 chance in $6^3 = 216$ to win.

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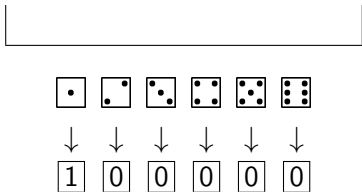
1 ticket	\$180	215 tickets	-\$1
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Classifying and Counting

Some chances processes involve **counting**. Box models can be used to describe such processes if the boxes are set up correctly.

Example

The number of aces in 50 rolls of a fair die is like the sum of _____ draws made at random with replacement from the box



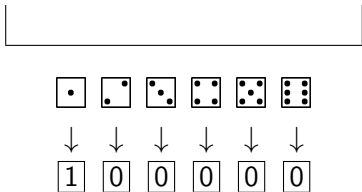
If you have to classify and count the draws, put 0's and 1's on the tickets. Mark 1 on the tickets that count for you, 0 on the others.

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1 ticket 1 5 tickets 0

					
↓	↓	↓	↓	↓	↓
1	0	0	0	0	0

If you have to classify and count the draws, put 0's and 1's on the tickets. Mark 1 on the tickets that count for you, 0 on the others.

Example: Number of Democrats

A town has 30,000 registered voters. An investigator want to know the percentage of Democrats in this town. A random sample of 1,000 registered voters will be drawn from the population.

some 0's some 1's

- ▶ Each registered voter in the town has one ticket in the box
- ▶ The tickets for Democrats should be marked 1 and the other tickets as 0.
- ▶ So there should be _____ tickets in the box.
- ▶ The number of Democrats in the sample is like the sum of _____ draws made at random _____ replacement from the box.

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- ▶ The tickets for Democrats should be marked 1 and the other tickets as 0.
- ▶ So there should be 30,000 tickets in the box.
- ▶ The number of Democrats in the sample is like the sum of 1,000 draws made at random without replacement from the box.